

Bridging the Quantum Divide: A Strategic Roadmap for Quantum Computing Adoption in a Developing Economy (Sri Lanka)

ABSTRACT

Quantum computing (QC) represents a fundamental paradigm shift in computational power, promising exponential speedups for specialized problems in optimization, cryptography, material sciences, and machine learning. While developed nations are investing billions into local physical quantum processors, developing economies like Sri Lanka face deep economic constraints, infrastructure deficits, and high intellectual brain drain, risking a widening technological chasm. This article provides a comprehensive situational analysis of Sri Lanka's high-performance computing readiness and outlines a practical, low-capital framework for rapid QC adoption. By leveraging cloud-based quantum architectures (Quantum-as-a-Service) combined with aggressive curricular re-engineering and public-private-academic collaborations, Sri Lanka can develop a highly skilled quantum-ready workforce. High-impact localized use cases—spanning maritime logistical routing at the Colombo Port, agricultural optimization, and macro-financial risk management—are analyzed to demonstrate immediate socioeconomic returns. Ultimately, this paper serves as an actionable blueprint for national policymakers, industrial leaders, and academic administrators to cultivate an agile quantum ecosystem without capital-intensive infrastructure outlays.

Keywords: Quantum Computing, Developing Economies, Strategic Technology Roadmap, Quantum-as-a-Service (QaaS), Technological Leapfrogging, Sri Lanka.

1. INTRODUCTION

The global computational landscape is on the precipice of a disruptive transformation driven by the transition from classical binary logic to quantum mechanics. Leveraging the twin principles of superposition and entanglement, quantum computers process information via quantum bits (qubits), allowing them to evaluate highly complex, multi-dimensional state spaces simultaneously. Tech conglomerates and global superpowers have entered a hyper-competitive race to achieve fault-tolerant quantum computing, with annual public and private investments exceeding tens of billions of dollars globally. However, this race risks leaving developing nations behind, widening the computational divide into an unbridgeable tech-gulf.

For a nation like Sri Lanka, which is actively seeking to revitalize its economy through digital transformation and high-value software exports, quantum readiness cannot be treated as a distant luxury. Historically, developing nations have suffered from delayed tech adoption cycles, often onboarding technologies only when they reach mass commoditization. Quantum computing demands a different strategy because the theoretical framework, algorithmic design, and cognitive models required to build quantum applications differ entirely from classical programming languages. If a nation lacks a workforce trained in quantum algorithms, the presence of affordable hardware later will be useless.

This paper shifts the discourse from physical infrastructure manufacturing to cloud-based algorithmic empowerment. We explore how Sri Lanka can bypass the capital-intensive phase of constructing physical Quantum Processing Units (QPUs) by strategically tapping into global cloud networks. By addressing the core architectural pillars of education, public-private alliances, and contextual policy development, Sri Lanka can achieve an agile, competitive stance in the upcoming quantum era.

2. CURRENT LANDSCAPE AND SITUATIONAL ANALYSIS

To effectively position Sri Lanka within the global quantum matrix, a realistic assessment of its current high-performance computing (HPC) and information communication technology (ICT) ecosystems is required. Sri Lanka possesses a vibrant and globally recognized ICT export sector, contributing significantly to its GDP via sophisticated software products and outsourced services. However, its academic and physical infrastructure is primarily engineered for classical software paradigms.

The primary barrier to local adoption is the high capital cost associated with quantum infrastructure. Physical QPUs require specialized cryogenic systems to maintain operating environments near absolute zero ($T \approx 15 \text{ mK}$), isolating fragile qubits from environmental decoherence. For a developing nation navigating macroeconomic restructuring, allocating capital for such infrastructure is impossible. Therefore, the strategic paradigm must shift towards utilizing cloud-hosted quantum backends, such as IBM Quantum, Amazon Braket, or Microsoft Azure Quantum.

The secondary gap lies within academia. While local state universities (e.g., Colombo, Peradeniya, Moratuwa) offer exceptional core programs in theoretical physics and pure mathematics, these domains are rarely cross-pollinated with computer science engineering. Consequently, students graduate with excellent programming proficiencies but little to no exposure to quantum mechanics or quantum information systems (QIS).

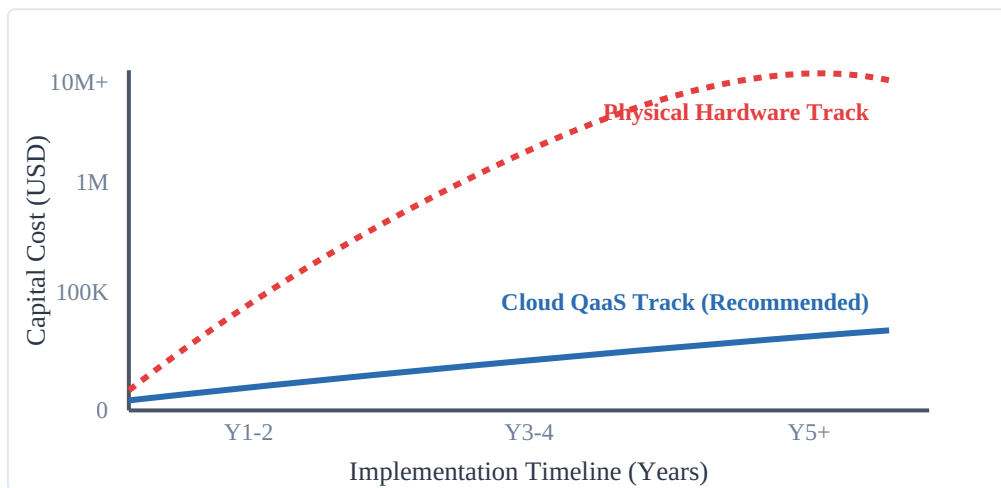


Figure 1: Comparative cost analysis between local physical hardware acquisition and a Cloud-Based Quantum-as-a-Service (QaaS) strategy over a 5-year macro-horizon.

3. HIGH-IMPACT LOCALIZED USE CASES

To secure state interest and industrial funding, quantum computing research must be tied to high-priority sectors. Sri Lanka can focus on areas where quantum algorithms offer significant optimization improvements over classical approaches.

Table 1: Strategic Sectors, Quantum Implementations, and Specific Economic Targets for Sri Lanka.

Economic Sector	Core Quantum Methodology	Localized Sri Lankan Context & Utility
Maritime & Logistics	Quantum Approximate Optimization Algorithm (QAOA) and Quantum Annealing for multi-variable routing.	Optimizing transshipment schedules, container placement, and anchorage allocation at the Port of Colombo to combat regional competition.
Agriculture	Variational Quantum Eigensolver (VQE) for molecular electronic structure simulation.	Simulating complex chemical nitrogen-fixation processes to develop cheaper local organic/chemical fertilizers and forecasting localized multi-crop yields.
Finance & Banking	Quantum Amplitude Estimation (QAE) for Monte Carlo simulations and portfolio risk management.	Protecting national banking infrastructure from international volatility, optimizing liquidity reserves, and enhancing macro-economic modeling.
Energy Grid	Quantum Integer Linear Programming for decentralized distribution.	Balancing fluctuating solar and wind inputs into the Ceylon Electricity Board (CEB) national grid, minimizing transmission line losses.

3.1. Case Example: Maritime Logistics Optimization

Consider the Port of Colombo, a vital maritime hub in South Asia. Port logistics suffer from the combinatorial explosion problem, where scheduling N ships across M berths with varying crane capacities becomes too complex for classical computers to optimize in real-time. This can be mathematically framed as a Quadratic Unconstrained Binary Optimization (QUBO) problem, which maps directly onto quantum annealing hardware or gate-based QAOA architectures. The cost function can be modeled as:

$$H(x) = \sum_i c_i x_i + \sum_{i < j} q_{ij} x_i x_j$$

where $x_i \in \{0, 1\}$ represent binary operational choices (e.g., ship berth assignments), c_i represents linear cost parameters, and q_{ij} captures interactive penalties (e.g., berthing delays, resource conflicts). Resolving Equation 1 via quantum hardware minimizes ship turnaround times, directly enhancing port efficiency and regional competitiveness.

4. THE THREE-PILLAR ROADMAP

We propose an integrated, phase-governed strategy categorized into three foundational pillars spanning a 5-year timeline. This structure targets human capital first, followed by commercialization, and concludes with institutional policy design.



Figure 2: Milestone timeline and interdependencies of the recommended National Quantum Strategy.

4.1. Pillar A: Academic Integration and Open-Access Cloud Sandboxes

The immediate requirement is to integrate Quantum Information Science (QIS) concepts into existing university undergraduate tracks. Computer science degrees should introduce elective courses in quantum algorithms, teaching students how to use open-source frameworks like IBM's Qiskit, Google's Cirq, or Rigetti's Forest. Rather than learning abstract mathematics, students should actively compile and run software on cloud-connected physical quantum backends via free tiers for researchers.

4.2. Pillar B: Public-Private-Academic Alliances

During the intermediate phase, a cohesive bridge must link research with commercial industry. A national body, the "Sri Lanka Quantum Technology Alliance" (SLQTA), should be established. This body would pair academic researchers with local tech players (e.g., Dialog Axiata, SLT-Mobitel, Virtusa, WSO2) to build sandbox

optimization systems. By framing business issues as quantum challenges, the industry can fund research initiatives in exchange for future commercial applications.

4.3. Pillar C: Policy Framework and Post-Quantum Cryptography

As quantum hardware scales globally, modern cryptographic standards (e.g., RSA and ECC) will become vulnerable to Shor's algorithm. Consequently, the final pillar emphasizes national security and policy development. The Sri Lanka Computer Emergency Readiness Team (SLCERT), alongside telecommunications authorities, must draft a roadmap for Post-Quantum Cryptography (PQC). This policy must mandate the gradual migration of public financial infrastructure, state databases, and military communications to quantum-resistant lattice-based cryptographic standards.

5. CHALLENGES AND STRATEGIC MITIGATIONS

Implementing this roadmap introduces several risks that require proactive management:

The Brain Drain Factor: Sri Lanka faces a persistent challenge with high intellectual brain drain, as local software engineering and physics graduates frequently seek opportunities abroad. To counter this, local initiatives must secure remote research grants and establish specialized tech roles within the private sector, allowing local talent to work on advanced international projects from within Sri Lanka.

Geopolitical Cloud Vulnerabilities: Relying entirely on cloud-based ecosystems exposes national computational tasks to foreign infrastructure controls and sanctions. To mitigate this risk, Sri Lanka should build localized high-performance classical clusters running quantum simulators, ensuring critical workflows can operate independently during external supply issues.

6. CONCLUSION

Sri Lanka can participate in the ongoing quantum revolution without investing in capital-intensive physical computing facilities. By adopting a software-centric, cloud-enabled strategy, the nation can build a specialized workforce ready to apply quantum methodologies to major sectors like maritime logistics, agriculture, and finance. Acting early on these educational and policy initiatives will allow Sri Lanka to develop an agile tech economy, secure its digital infrastructure against future threats, and establish itself as a key hub for quantum engineering in the region.

References

- [1] Arute, F., Arya, K., Babbush, R., et al. (2019). Quantum supremacy using a programmable superconducting processor. *Nature*, 574(7779), 505-510.
- [2] Farhi, E., Goldstone, J., & Gutmann, S. (2014). A quantum approximate optimization algorithm. *arXiv preprint arXiv:1411.4028*.
- [3] National Institute of Standards and Technology (NIST). (2024). Post-Quantum Cryptography Standardization Roadmap. *NIST Special Publications*.

- [4] Information and Communication Technology Agency (ICTA) of Sri Lanka. (2025). National Digital Transformation Strategy Framework. Official Report.
- [5] Gill, S. S., Garraghan, P., & Buyya, R. (2024). Quantum Computing as a Service (QaaS): State-of-the-art and future architectural directions. *Software: Practice and Experience*, 54(3), 412-431.